

CBSE CLASS 10 SCIENCE – COMPLETE REVISION NOTES

BIOLOGY (UNIT II: WORLD OF LIVING)

CHAPTER 1: LIFE PROCESSES

1. What are Life Processes?

The basic functions performed by living organisms to maintain life are called life processes.

****Types of Life Processes:****

- ****Nutrition:**** Obtaining and utilising food.
- ****Respiration:**** Releasing energy from food.
- ****Transport:**** Movement of substances.
- ****Excretion:**** Removing waste products.
- ****Reproduction:**** Producing offspring.

2. Nutrition

The process of obtaining and utilising food for energy, growth, and repair.

****Types of Nutrition:****

****A. Autotrophic Nutrition:****

- Organisms synthesise their own food from inorganic sources.
- ****Examples:**** Green plants, some bacteria.
- ****Photosynthesis Equation:**** $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow (\text{Sunlight, Chlorophyll}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$

****B. Heterotrophic Nutrition:****

- Organisms depend on other organisms for food.
- ****Types:****
 - ****Holozoic:**** Ingestion of solid food. ****Examples:**** Humans, amoeba, animals.
 - ****Saprotrophic:**** Feeding on dead and decaying organic matter. ****Examples:**** Fungi, bacteria.
 - ****Parasitic:**** Feeding on living organisms (hosts). ****Examples:**** Tapeworm, lice, cuscuta.

****Nutrition in Amoeba:****

1. ****Ingestion:**** Pseudopodia engulf the food particle.
2. ****Digestion:**** Food vacuole forms; digestive enzymes break down food.
3. ****Absorption:**** Nutrients absorbed into cytoplasm.
4. ****Assimilation:**** Nutrients used for energy and growth.
5. ****Egestion:**** Undigested waste expelled.

****Nutrition in Human Beings:****

1. ****Ingestion:**** Taking food into the mouth.
2. ****Digestion:**** Breaking down food.
3. ****Absorption:**** Nutrients absorbed in small intestine.
4. ****Assimilation:**** Nutrients used by cells.
5. ****Egestion:**** Elimination of undigested waste.

3. Human Digestive System

****A. Mouth:****

- ****Teeth:**** Mechanical digestion (chewing).
- ****Salivary Glands:**** Secrete saliva containing salivary amylase.
- ****Salivary Amylase:**** Starch → Maltose (partial digestion).
- ****pH:**** Saliva has pH ~6.8 (slightly acidic).

****B. Oesophagus (Food Pipe):****

- Peristalsis (wave-like muscle contractions) pushes food down.
- No digestion occurs here.

****C. Stomach:****

- ****Gastric Glands:**** Secrete gastric juice.
- ****Components of Gastric Juice:****
 - ****Pepsinogen → (HCl) → Pepsin:**** Proteins → Peptides.
 - ****Hydrochloric Acid (HCl):**** Kills bacteria; provides acidic medium for pepsin.
 - ****Mucus:**** Protects stomach wall from HCl.
- ****Chyme:**** Semi-digested food that passes to the small intestine.

****D. Small Intestine:****

- Long, coiled tube (about 6 metres).
- ****Parts:**** Duodenum, Jejunum, Ileum.
- ****Secretions:****
 - ****Bile (from Liver):**** Emulsifies fats (breaks into small droplets). Makes medium alkaline.
 - ****Pancreatic Juice (from Pancreas):****
 - ****Trypsin:**** Proteins → Peptides.

- **Lipase:** Fats → Fatty acids + Glycerol.
- **Amylase:** Starch → Maltose.
- **Intestinal Juice (Succus Entericus):** Completes digestion of carbohydrates, proteins, and fats.
- **Final Products:** Carbohydrates → Glucose; Proteins → Amino acids; Fats → Fatty acids + Glycerol.
- **Absorption:** Villi (finger-like projections) increase surface area for absorption.

E. Large Intestine:

- Absorbs water and minerals from undigested food.
- **Rectum:** Stores faeces temporarily.
- **Anus:** Elimination of faeces (egestion).

F. Accessory Organs:

- **Liver:** Largest gland; produces bile (stored in gall bladder).
- **Gall Bladder:** Stores and concentrates bile.
- **Pancreas:** Produces pancreatic juice (amylase, trypsin, lipase).

4. Respiration

The process of releasing energy from food (glucose).

A. Types of Respiration:

Aerobic Respiration:

- **Oxygen:** Required.
- **End products:** $\text{CO}_2 + \text{H}_2\text{O}$.
- **Energy released:** 38 ATP (more).
- **Occurs in:** Mitochondria.
- **Equation:** $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy (38 ATP)}$

Anaerobic Respiration:

- **Oxygen:** Not required.
- **End products:** Ethanol + CO_2 (in plants) or Lactic acid (in animals).
- **Energy released:** 2 ATP (less).
- **Occurs in:** Cytoplasm.
- **Equations:**
 - **Yeast:** $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2 + \text{Energy (2 ATP)}$
 - **Muscle cells:** $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_3\text{H}_6\text{O}_3 \text{ (Lactic acid)} + \text{Energy (2 ATP)}$

B. Human Respiratory System:

****Parts:****

1. ****Nasal Cavity:**** Filters, warms, and moistens air.
2. ****Pharynx:**** Common passage for air and food.
3. ****Larynx:**** Voice box (vocal cords).
4. ****Trachea:**** Windpipe; rings of cartilage prevent collapse.
5. ****Bronchi:**** Trachea divides into left and right bronchi.
6. ****Bronchioles:**** Smaller tubes branching from bronchi.
7. ****Alveoli:**** Air sacs at the end of bronchioles; site of gas exchange.
 - ****Structure:**** Thin walls (one cell thick), surrounded by blood capillaries.
 - ****Function:**** Exchange of O₂ and CO₂ by diffusion.

****Mechanism of Breathing:****

- ****Inhalation (Inspiration):****
 - Diaphragm contracts (flattens).
 - Rib muscles contract (ribs move up and out).
 - Chest cavity volume increases.
 - Air pressure decreases; air rushes in.
- ****Exhalation (Expiration):****
 - Diaphragm relaxes (becomes dome-shaped).
 - Rib muscles relax.
 - Chest cavity volume decreases.
 - Air pressure increases; air rushes out.

****Transport of Gases:****

- ****Oxygen:**** 97% transported by haemoglobin in RBCs; 3% dissolved in plasma.
 - $\text{Hb} + \text{O}_2 \rightarrow \text{HbO}_2$ (Oxyhaemoglobin).
- ****Carbon Dioxide:**** 70% carried as bicarbonate ions; 20-25% by haemoglobin; 7% dissolved in plasma.

5. Transportation

****A. Transportation in Plants:****

****Xylem:****

- ****Function:**** Transports water and minerals from roots to leaves.
- ****Components:**** Tracheids, vessels, xylem fibres, xylem parenchyma.
- ****Mechanism:****
 1. ****Root Pressure:**** Water enters roots by osmosis.
 2. ****Cohesion-Tension Theory (Transpiration Pull):****
 - ****Transpiration:**** Loss of water vapour from leaves creates suction.
 - ****Cohesion:**** Water molecules stick to each other.
 - ****Adhesion:**** Water molecules stick to xylem walls.

- Forms a continuous water column pulled up.

****Phloem:****

- ****Function:**** Transports food (sugars) from leaves to other parts.
- ****Components:**** Sieve tubes, companion cells, phloem fibres, phloem parenchyma.
- ****Mechanism (Pressure Flow Hypothesis):****
 - ****Source (Leaves):**** Sugar actively loaded into sieve tubes; water enters → high pressure.
 - ****Sink (Roots/Fruits):**** Sugar unloaded; water leaves → low pressure.
- Flow occurs from high pressure to low pressure.

****B. Transportation in Animals:****

****Blood Components:****

- ****Plasma (55%):**** Liquid matrix; contains water, proteins, glucose, salts, hormones.
- ****RBCs (Erythrocytes):**** Transport O₂; contain haemoglobin.
- ****WBCs (Leucocytes):**** Defend against infections.
- ****Platelets (Thrombocytes):**** Help in blood clotting.

****Blood Vessels:****

- ****Arteries:**** Carry blood away from the heart; thick, elastic walls; high pressure; no valves.
- ****Veins:**** Carry blood to the heart; thin walls; low pressure; valves present.
- ****Capillaries:**** Connect arteries and veins; walls one cell thick; exchange of materials.

****Heart:****

- Four-chambered organ (two atria, two ventricles).

****Double Circulation in Humans:****

1. ****Pulmonary Circulation:**** Right ventricle → Pulmonary artery → Lungs → Pulmonary veins → Left atrium.
2. ****Systemic Circulation:**** Left ventricle → Aorta → Body tissues → Vena cava → Right atrium.

****Valves:**** Prevent backflow of blood.

- ****Tricuspid valve:**** Between right atrium and right ventricle.
- ****Bicuspid (Mitral) valve:**** Between left atrium and left ventricle.
- ****Semilunar valves:**** At the base of arteries.

****Lymph:****

- ****Function:**** Returns tissue fluid to blood; transports fats; defends against infection.
- ****Lymph Nodes:**** Filter lymph; contain WBCs.

6. Excretion

Removal of metabolic waste products from the body.

****A. Excretion in Humans:****

****Kidneys:****

- ****Location:**** At the back of the abdominal cavity.
- ****Functions:**** Filter blood; produce urine.
- ****Structure:**** Outer cortex, inner medulla, pelvis.

****Nephron (Functional Unit):****

1. ****Bowman's Capsule + Glomerulus = Malpighian Corpuscle.****
 - ****Ultrafiltration:**** Blood pressure forces water, salts, glucose, urea, amino acids into the capsule.
 - Large molecules (proteins, blood cells) remain in blood.
2. ****Proximal Convoluted Tubule (PCT):****
 - Reabsorption: Glucose, amino acids, most water, useful salts.
3. ****Loop of Henle:**** Reabsorption of water and sodium chloride.
4. ****Distal Convoluted Tubule (DCT):****
 - Tubular secretion (removal of additional wastes).
 - Further reabsorption of water.
5. ****Collecting Duct:**** Urine collected and sent to the pelvis.

****Urine Composition:****

- ****Normal:**** 95% water; 5% urea, uric acid, salts, creatinine.
- ****Abnormal:**** Glucose → Diabetes mellitus; Proteins → Kidney damage.

****Ureters, Urinary Bladder, Urethra:****

- ****Ureter:**** Carries urine from kidney to bladder.
- ****Urinary Bladder:**** Stores urine.
- ****Urethra:**** Carries urine out of the body.

****B. Excretion in Plants:****

- ****Gaseous Wastes:**** CO₂ (via stomata), O₂ (via stomata).
- ****Solid Wastes:**** Resins, gums, tannins, alkaloids (stored in vacuoles, bark, leaves).
- ****Excess Water:**** Transpiration (water vapour), Guttation (liquid water through hydathodes).

IMPORTANT QUESTIONS

****Very Short Answer (1-2 marks):****

1. What is the role of saliva in the digestion of food?
2. Name the enzyme that converts starch into maltose.
3. What is the function of villi in the small intestine?

4. Define photosynthesis. Write the equation.
5. What is the difference between autotrophic and heterotrophic nutrition?
6. Name the site of aerobic and anaerobic respiration.

****Short Answer (3 marks):****

1. Explain the process of nutrition in Amoeba.
2. Describe the structure and function of nephron.
3. Explain the mechanism of breathing in humans.
4. How is oxygen transported in the blood?
5. Differentiate between xylem and phloem.

****Long Answer (5 marks):****

1. Describe the human digestive system with a labelled diagram.
2. Explain the process of photosynthesis with the equation. What is the significance of photosynthesis?
3. Describe the structure and working of the human heart.
4. What is the role of kidneys in excretion? Explain the structure and function of a nephron.
5. Compare aerobic and anaerobic respiration. Write the equations for both.

****Case-Based Question:****

Read the following and answer:

"A student conducted an experiment to show that CO_2 is given out during respiration. He took germinating seeds in a flask with KOH solution. The KOH solution absorbed CO_2 , creating a vacuum."

- (i) What is the purpose of KOH in the experiment?
- (ii) What observation would he make?
- (iii) Why do germinating seeds release CO_2 ?
- (iv) Write the equation for aerobic respiration.
- (v) What would happen if KOH is not used?

CHAPTER 2: CONTROL AND COORDINATION

1. Control and Coordination in Animals

****A. Nervous System:****

- ****Neuron:**** Structural and functional unit.
- ****Structure:****
 - ****Dendrites:**** Receive impulses.
 - ****Cell Body (Cyton):**** Contains nucleus; integrates impulses.

- **Axon:** Long fibre; conducts impulses away from cell body.
- **Myelin Sheath:** Insulating layer; speeds up impulse conduction.
- **Synapse:** Gap between neurons; chemical (neurotransmitters) transmission.

B. Types of Nervous System:

1. **Central Nervous System (CNS):** Brain + Spinal cord.
2. **Peripheral Nervous System (PNS):** Nerves connecting CNS to body parts.

C. Human Brain:

1. **Forebrain (Cerebrum):**
 - Largest part; divided into two hemispheres.
 - **Functions:** Thinking, memory, intelligence, reasoning, voluntary actions, sensory perception.
2. **Midbrain:**
 - Connects forebrain to hindbrain.
 - Controls reflex movements of eyes and head.
3. **Hindbrain:**
 - **Cerebellum:** Balance, coordination of voluntary movements.
 - **Medulla Oblongata:** Involuntary actions (breathing, heartbeat, digestion, blood pressure).
 - **Pons:** Relay station between cerebrum and medulla.

D. Spinal Cord:

- Extends from medulla oblongata.
- **Functions:** Conducts impulses to and from the brain; centre for reflex actions.

E. Reflex Action:

- Rapid, automatic, involuntary response.
- **Reflex Arc:** Receptor → Sensory neuron → Spinal cord (Relay neuron) → Motor neuron → Effector.
- **Examples:** Knee-jerk, blinking, withdrawing hand from hot object, sneezing.

F. Nervous vs Endocrine System:

Nervous System:

- **Messages:** Electrical impulses.
- **Speed:** Very fast.
- **Duration:** Short-lived.
- **Pathway:** Via neurons.
- **Target:** Specific organs/cells.

Endocrine System:

- **Messages:** Chemical messages (hormones).
- **Speed:** Slow.
- **Duration:** Long-lasting.

- **Pathway:** Via blood.
- **Target:** Whole body (target organs).

2. Coordination in Plants

A. Plant Hormones (Phytohormones):

Auxin:

- **Functions:** Cell elongation; apical dominance; root initiation; promotes phototropism and geotropism (shoots).

Gibberellin:

- **Functions:** Stem elongation; seed germination; bolting (rapid stem growth); flowering.

Cytokinin:

- **Functions:** Cell division; delays ageing (senescence); promotes lateral bud growth.

Absciscic Acid (ABA):

- **Functions:** Growth inhibitor; induces dormancy; stomatal closure (stress hormone); abscission (falling of leaves).

Ethylene:

- **Functions:** Fruit ripening; senescence; abscission.

B. Tropic Movements:

Directional growth movements in response to external stimuli.

Phototropism:

- **Stimulus:** Light.
- **Shoot Response:** Positive (towards light).
- **Root Response:** Negative (away from light).

Geotropism:

- **Stimulus:** Gravity.
- **Shoot Response:** Negative (away from gravity).
- **Root Response:** Positive (towards gravity).

Hydrotropism:

- **Stimulus:** Water.
- **Root Response:** Positive (towards water).

Chemotropism:

- **Stimulus:** Chemicals.
- **Example:** Pollen tube growth towards ovule.

Thigmotropism:

- **Stimulus:** Touch.
- **Shoot Response:** Positive (tendrils coil around support).

Mechanism of Tropic Movements:

- **Phototropism:** Light causes auxin to accumulate on the shaded side, causing cells there to elongate more, bending the shoot towards light.
- **Geotropism:** Gravity causes auxin to accumulate on the lower side, causing shoots to bend upwards and roots downwards.

3. Hormones in Animals

A. Endocrine Glands and Their Hormones:

Pituitary (Master Gland):

- **Growth Hormone (GH):** Growth of body.
- **TSH:** Stimulates thyroid.
- **ACTH:** Stimulates adrenal.
- **FSH/LH:** Stimulates gonads.
- **ADH (Vasopressin):** Water reabsorption in kidneys.
- **Oxytocin:** Uterine contractions; milk ejection.

Thyroid:

- **Thyroxine:** Regulates metabolism; growth and development.

Parathyroid:

- **Parathormone (PTH):** Calcium and phosphate balance.

Adrenal:

- **Adrenaline:** "Fight or flight" response; increases heart rate, blood pressure, glucose.
- **Cortisol:** Stress response; metabolism.

Pancreas:

- **Insulin:** Lowers blood glucose.
- **Glucagon:** Raises blood glucose.

Gonads (Testes):

- **Testosterone:** Male secondary sexual characters; sperm production.

****Gonads (Ovaries):****

- ****Oestrogen:**** Female secondary sexual characters; menstrual cycle.
- ****Progesterone:**** Prepares/maintains uterus for pregnancy.

****Thymus:****

- ****Thymosin:**** Maturation of T-lymphocytes.

****Pineal:****

- ****Melatonin:**** Sleep-wake cycle (circadian rhythm).

****B. Feedback Mechanism (Negative Feedback):****

- Maintains hormonal balance.
- ****Example:**** High blood glucose → Insulin secretion → Glucose converted to glycogen → Blood glucose decreases → Insulin secretion stops.

4. Chemical Coordination in Animals

****Hormones:****

- Chemical messengers secreted by endocrine glands.
- Transported by blood.
- Act on specific target organs.
- Effects are slow but long-lasting.

****Adrenaline (Emergency Hormone):****

- ****Source:**** Adrenal medulla.
- ****Effects:**** Increases heart rate, blood pressure, breathing rate; diverts blood to muscles; increases blood glucose.
- ****Purpose:**** Prepares body for "fight or flight".

****Thyroxine:****

- ****Source:**** Thyroid gland.
- ****Effects:**** Increases metabolic rate; growth and development.
- ****Deficiency (Hypothyroidism):****
 - ****In adults:**** Myxoedema (swelling, lethargy, weight gain).
 - ****In children:**** Cretinism (stunted growth, mental retardation).
 - ****Goitre:**** Swelling of thyroid due to iodine deficiency.
- ****Excess (Hyperthyroidism):**** Exophthalmic goitre (bulging eyes, weight loss, irritability).

****Insulin and Glucagon:****

- ****Insulin (Beta cells):**** Decreases blood glucose.
- ****Glucagon (Alpha cells):**** Increases blood glucose.

- **Diabetes Mellitus:** Lack of insulin → High blood sugar (glucose in urine). Treated with insulin injections.

IMPORTANT QUESTIONS

Very Short Answer (1-2 marks):

1. Define reflex action. Give an example.
2. What is the role of auxin in phototropism?
3. Name the hormone that regulates blood sugar level.
4. What is the function of the cerebellum?
5. Differentiate between tropic and nastic movements.

Short Answer (3 marks):

1. Draw a labelled diagram of a neuron. Explain its structure.
2. Describe the structure and functions of the human brain.
3. Explain the reflex arc with an example.
4. What are plant hormones? Name and describe the functions of any four.
5. Explain the feedback mechanism with an example.

Long Answer (5 marks):

1. Compare and contrast nervous coordination and chemical coordination in animals.
2. Describe the major endocrine glands, their hormones, and functions.
3. Explain the mechanism of tropic movements in plants.
4. What is the role of hormones in the human body? Explain with examples.
5. Describe the structure of the human brain and the functions of its different parts.

Case-Based Question:

Read the following and answer:

"A person accidentally touches a hot pan and immediately withdraws his hand."

- (i) What type of action is this?
- (ii) Trace the pathway of the nerve impulse (reflex arc).
- (iii) Name the organ involved in this reflex action.
- (iv) Would this action occur if the spinal cord was damaged? Why?
- (v) What is the significance of such reflexes?

CHAPTER 3: REPRODUCTION

1. Reproduction – Basic Concepts

- **Reproduction:** The biological process by which organisms produce new individuals of the same species.

- **Types:**

- **Asexual Reproduction:** Single parent; no gametes; offspring identical to parent.
- **Sexual Reproduction:** Two parents; gametes fuse; offspring show variation.

Significance of Reproduction:

- Ensures continuity of species.
- Maintains population.
- Variations allow evolution and adaptation.

2. Asexual Reproduction

A. Fission:

- **Binary Fission:** Parent divides into two equal halves.
 - **Amoeba:** Splits into two.
 - **Leishmania:** Longitudinal binary fission.
 - **Paramecium:** Transverse binary fission.
- **Multiple Fission:** Parent divides into many daughter cells.
 - **Example:** Plasmodium (malarial parasite).

B. Budding:

- **Example:** Yeast, Hydra.
- A small outgrowth (bud) develops on the parent and eventually detaches to form a new individual.

C. Fragmentation:

- **Example:** Spirogyra (green alga), Planaria.
- The organism breaks into fragments, each capable of growing into a new individual.

D. Regeneration:

- **Example:** Hydra, Planaria, starfish.
- The organism can regrow lost body parts.

E. Spore Formation:

- **Example:** Rhizopus (bread mould), Mucor.
- Spores are produced in sporangia; they are resistant and can survive unfavourable conditions.

F. Vegetative Propagation:

- **Definition:** Asexual reproduction in plants using vegetative parts.

- **Methods:**
 - **Stem:** Potato (eyes), ginger (rhizome), sugarcane.
 - **Root:** Sweet potato, dahlia, carrot.
 - **Leaves:** Bryophyllum (leaf buds).
 - **Bulbs:** Onion, garlic.
- **Advantages:**
 - Faster and cheaper.
 - Plants are identical to parent (desirable traits preserved).
 - Can produce seedless varieties.
- **Disadvantages:** No new varieties; susceptible to diseases.

G. Tissue Culture (Micropropagation):

- Growing plants from small tissue samples in a nutrient medium.
- **Advantages:** Produces many plants in a short time; disease-free plants.

3. Sexual Reproduction in Flowering Plants

A. Flower Structure:

- **Androecium (Male):**
 - **Stamen:** Anther (produces pollen grains containing male gametes) + Filament.
- **Gynoecium (Female):**
 - **Carpel/Pistil:** Stigma (receives pollen), Style (tube), Ovary (contains ovules with female gametes).

B. Pollination:

- Transfer of pollen grains from anther to stigma.
- **Types:**
 - **Self-Pollination:** Same flower or same plant.
 - **Cross-Pollination:** Different plants.
- **Agents:** Wind, water, insects, birds, animals.
- **Adaptations for Insect Pollination:** Brightly coloured petals, nectar, fragrance.
- **Adaptations for Wind Pollination:** Light, dry pollen; large, feathery stigma.

C. Fertilisation:

1. Pollen grain lands on stigma.
2. Pollen tube grows down the style.
3. Pollen tube enters the ovule through the micropyle.
4. Two male gametes are released (double fertilisation).
5. One male gamete fuses with the egg cell (zygote formation).
6. The other male gamete fuses with the polar nuclei (endosperm formation).

D. Seed and Fruit Formation:

- **Ovule → Seed.**
- **Ovary → Fruit.**
- **Zygote → Embryo.**
- **Endosperm → Food storage.**

E. Seed Dispersal:

- **Agents:** Wind, water, animals, explosive mechanisms.
- **Adaptations:**
 - **Wind:** Light seeds with wings/parachutes.
 - **Water:** Floating seeds.
 - **Animals:** Sticky seeds or edible fruits.

4. Sexual Reproduction in Humans

A. Male Reproductive System:

1. **Testes (Paired):** Produce sperms and testosterone; located in scrotum.
2. **Epididymis:** Sperms mature and stored here.
3. **Vas Deferens (Sperm Duct):** Carries sperms.
4. **Seminal Vesicles, Prostate Gland, Cowper's Glands:** Secrete fluid (semen).
5. **Urethra:** Common passage for semen and urine.
6. **Penis:** Copulatory organ.

B. Female Reproductive System:

1. **Ovaries (Paired):** Produce ova and hormones (oestrogen, progesterone).
2. **Fallopian Tubes (Oviducts):** Transport ovum; site of fertilisation.
3. **Uterus (Womb):** Site of implantation and foetal development.
4. **Cervix:** Lower part of uterus.
5. **Vagina:** Birth canal; receives penis during copulation.

C. Fertilisation and Development:

1. **Fertilisation:** Fusion of sperm and ovum in the fallopian tube → Zygote.
2. **Implantation:** Zygote divides and implants in the uterine lining.
3. **Embryo Development:** Embryo (up to 8 weeks); Foetus (9 weeks onwards).
4. **Placenta:** Functions: Nutrition, respiration, excretion, hormone production.
5. **Gestation (Pregnancy):** Lasts about 9 months (40 weeks).
6. **Parturition:** Childbirth.

D. Menstrual Cycle:

- **Duration:** About 28 days.
- **Phases:**
 1. **Menstrual Phase (Days 1-5):** Endometrium sheds; bleeding occurs.
 2. **Follicular Phase (Days 6-13):** Follicle matures; oestrogen thickens endometrium.

3. **Ovulation (Day 14):** Release of ovum from ovary.
4. **Luteal Phase (Days 15-28):** Corpus luteum secretes progesterone.
5. If fertilisation does not occur → corpus luteum degenerates → menstruation.

E. Reproductive Health:

- **Family Planning:** Controlling number and spacing of children.
- **Methods of Contraception:**
 - **Barrier Methods:** Condoms, diaphragms.
 - **Chemical Methods:** Oral pills.
 - **Intrauterine Devices (IUDs):** Copper T.
 - **Surgical Methods:** Vasectomy (male), Tubectomy (female).
 - **Natural Methods:** Withdrawal, rhythm method.
- **Safe Sex vs HIV/AIDS:**
 - **AIDS:** Caused by HIV.
 - **Transmission:** Sexual contact, contaminated blood, sharing needles, mother to baby.
 - **Prevention:** Safe sex, screening blood, using sterile needles.

IMPORTANT QUESTIONS

Very Short Answer (1-2 marks):

1. Define reproduction. What are its types?
2. What is vegetative propagation? Give two examples.
3. Name the male and female gametes in humans.
4. What is the function of the placenta?
5. Define pollination. What are its types?

Short Answer (3 marks):

1. Explain any three methods of asexual reproduction with examples.
2. Describe the structure of a flower and its reproductive parts.
3. Explain the process of fertilisation in flowering plants.
4. Draw a labelled diagram of the male/female reproductive system.
5. What are the methods of contraception? Explain any two.

Long Answer (5 marks):

1. Describe the process of sexual reproduction in humans.
2. Explain the menstrual cycle in detail.
3. Describe the reproductive system of flowering plants with a diagram.
4. What is the importance of reproduction? Explain the various methods of contraception.
5. Compare sexual and asexual reproduction. Give examples.

Case-Based Question:

Read the following and answer:

"A couple wants to have children but is concerned about family planning. They visit a doctor for advice."

- (i) What are the different methods of contraception available?
- (ii) What is the role of the placenta during pregnancy?
- (iii) What is the importance of safe sex?
- (iv) How does HIV transmission occur?
- (v) Why is family planning important?

CHAPTER 4: HEREDITY AND EVOLUTION

1. Heredity

****Heredity:**** The transmission of traits from parents to offspring.

****Genes:**** Basic units of heredity; segments of DNA that code for specific traits.

****Genetics:**** The study of heredity and variation.

****Variation:**** Differences in traits among individuals of the same species.

****Role of Variation:****

- Allows organisms to adapt to changing environments.
- Provides raw material for evolution.

2. Mendel's Laws of Inheritance

Gregor Mendel conducted experiments on garden pea plants.

****Why Pea Plants?:****

- Short generation time.
- Large number of offspring.
- Easy to cross-breed.
- Visible contrasting traits (7 pairs).

****Mendel's Experiments:****

****A. Monohybrid Cross (One Trait):****

- **Example:** Tall plant (TT) × Dwarf plant (tt).
- **F₁ Generation:** All tall (Tt). Tall trait is dominant; dwarf trait is recessive.
- **F₂ Generation (Selfing of F₁):** Tt × Tt.
 - **Phenotypic Ratio:** 3 Tall : 1 Dwarf.
 - **Genotypic Ratio:** 1 TT : 2 Tt : 1 tt.

B. Dihybrid Cross (Two Traits):

- **Example:** Round Yellow (RRYY) × Wrinkled Green (rryy).
- **F₁ Generation:** All Round Yellow (RrYy).
- **F₂ Generation (Selfing of F₁):** RrYy × RrYy.
 - **Phenotypic Ratio:** 9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green.

Mendel's Laws:

- Law of Dominance:**
 - When two contrasting alleles are present, one (dominant) expresses itself, while the other (recessive) is masked.
- Law of Segregation (Law of Purity of Gametes):**
 - The two alleles of a gene separate during gamete formation.
 - Each gamete receives only one allele.
- Law of Independent Assortment:**
 - When two or more traits are studied, the inheritance of one trait is independent of the other.
 - Applies only to genes on different chromosomes.

3. Important Genetic Terms

- **Gene:** Basic unit of heredity.
- **Allele:** Alternative forms of a gene.
- **Dominant Allele:** Allele that expresses itself in heterozygous condition.
- **Recessive Allele:** Allele whose expression is masked in heterozygous condition.
- **Homozygous:** Both alleles are identical.
- **Heterozygous:** Alleles are different.
- **Genotype:** Genetic constitution.
- **Phenotype:** Observable physical appearance.
- **Chromosome:** Thread-like structure containing DNA.
- **DNA:** Double helix molecule; stores genetic information.

4. Sex Determination

****In Humans:****

- ****Chromosomes:**** 23 pairs (22 pairs autosomes + 1 pair sex chromosomes).
- ****Females:**** XX (homogametic) → Produce X-bearing eggs.
- ****Males:**** XY (heterogametic) → Produce X-bearing and Y-bearing sperms.
- ****Determination of Sex:****
 - ****XX (Female):**** X sperm + X egg.
 - ****XY (Male):**** Y sperm + X egg.

****In Other Organisms:****

- ****Birds (ZW System):**** Female is ZW; male is ZZ.
- ****Insects (XO System):**** Female is XX; male is XO.
- ****Bees (Haplo-Diploid):**** Males are haploid; females are diploid.

5. Evolution

****Evolution:**** Gradual changes in organisms over generations, leading to the development of new species.

****A. Evidence for Evolution:****

1. ****Fossils:**** Preserved remains of ancient organisms; show sequence of development.
2. ****Homologous Organs:**** Same basic structure; different functions. Examples: Forelimbs of humans, whales, bats, birds. Indicate common ancestry.
3. ****Analogous Organs:**** Different basic structures; same functions. Examples: Wings of insects and birds. Indicate convergent evolution.
4. ****Vestigial Organs:**** Organs with no apparent function. Examples: Appendix in humans; wisdom teeth.
5. ****Comparative Embryology:**** Similar early embryonic development across species.
6. ****Molecular Biology:**** Similar DNA and proteins indicate common ancestry.

****B. Theories of Evolution:****

1. ****Lamarck's Theory (Inheritance of Acquired Characters):****
 - Proposed use and disuse of organs; acquired traits are inherited.
 - Example: Giraffe's long neck.
 - ****Criticism:**** Acquired traits are not inherited.
2. ****Darwin's Theory (Natural Selection):****
 - ****Key Points:****
 - ****Overproduction:**** More offspring than can survive.
 - ****Struggle for Existence:**** Competition for resources.

- **Variations:** Differences among individuals.
- **Natural Selection:** Individuals with advantageous variations survive and reproduce.
- **Survival of the Fittest:** Nature selects the best-adapted organisms.

C. Speciation:

- Formation of new species from existing populations.
- **Factors:**
 - **Geographical Isolation:** Separation by physical barriers.
 - **Reproductive Isolation:** Inability to interbreed.
 - **Genetic Drift:** Random changes in allele frequencies.
 - **Natural Selection:** Different environments select for different traits.

6. Human Evolution

- **Common Ancestor:** Humans share a common ancestor with chimpanzees and gorillas.
- **Fossil Evidence:**
 - **Australopithecus:** Early hominid (Lucy).
 - **Homo habilis:** "Handy man"; used tools.
 - **Homo erectus:** Walked upright; used fire.
 - **Homo sapiens:** Modern humans; developed culture, language, art.
- **Migration:** Humans originated in Africa and migrated to other continents.

IMPORTANT QUESTIONS

Very Short Answer (1-2 marks):

1. Define heredity and variation.
2. What is a gene?
3. State Mendel's Law of Dominance.
4. What is the difference between homologous and analogous organs?
5. What is speciation?
6. How is sex determined in humans?

Short Answer (3 marks):

1. Explain Mendel's monohybrid cross with a Punnett square.
2. What are the evidences for evolution? Explain any two.
3. Compare and contrast homologous and analogous organs with examples.
4. Describe Darwin's theory of natural selection.
5. How does geographical isolation lead to speciation?

Long Answer (5 marks):

1. Describe Mendel's laws of inheritance with examples.
2. Explain the dihybrid cross with a Punnett square.
3. Discuss the evidence for evolution with examples.
4. What is evolution? Explain the theories of Lamarck and Darwin.
5. Describe the factors that lead to speciation.

****Case-Based Question:****

Read the following and answer:

"In a pea plant experiment, Mendel crossed a tall plant (TT) with a dwarf plant (tt). All F₁ plants were tall. When F₁ plants were self-pollinated, the F₂ generation had both tall and dwarf plants."

- (i) What is the genotype of the F₁ generation?
- (ii) What is the phenotypic ratio of the F₂ generation?
- (iii) Why did all F₁ plants show the tall trait?
- (iv) What is the law demonstrated by this cross?
- (v) What is the difference between genotype and phenotype?

CHEMISTRY (UNIT I: CHEMICAL SUBSTANCES – NATURE AND BEHAVIOUR)

CHAPTER 5: CHEMICAL REACTIONS AND EQUATIONS

1. Chemical Reactions

A chemical reaction is a process in which one or more substances (reactants) are transformed into one or more new substances (products) with different properties.

****Indicators of a Chemical Reaction:****

- Change in state.
- Change in colour.
- Evolution of gas.
- Change in temperature.
- Formation of precipitate.

2. Chemical Equations

A chemical equation is a symbolic representation of a chemical reaction.

Word Equation: Magnesium + Oxygen → Magnesium oxide

Chemical Equation: $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$

Balanced Chemical Equation: A chemical equation in which the number of atoms of each element is the same on both sides.

Steps to Balance:

1. Write the unbalanced equation.
2. Compare the number of atoms of each element on both sides.
3. Use coefficients to balance atoms one element at a time.
4. Ensure the smallest whole number coefficients are used.
5. Check that all atoms are balanced.

Example: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$

- $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$

3. Types of Chemical Reactions

A. Combination Reaction:

Two or more reactants combine to form a single product.

- $\text{A} + \text{B} \rightarrow \text{AB}$

- **Examples:**

- $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$

- $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$

B. Decomposition Reaction:

A single compound breaks down into two or more simpler substances.

- $\text{AB} \rightarrow \text{A} + \text{B}$

- **Types:**

- **Thermal:** $2\text{Pb(NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$

- **Electrolytic:** $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

- **Photolytic:** $2\text{AgCl} \rightarrow 2\text{Ag} + \text{Cl}_2$

C. Displacement Reaction:

A more reactive element displaces a less reactive element from its compound.

- $\text{A} + \text{BC} \rightarrow \text{AC} + \text{B}$

- **Example:** $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$

D. Double Displacement Reaction:

Two compounds exchange their ions to form two new compounds.

- $AB + CD \rightarrow AD + CB$
- **Example:** $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

E. Precipitation Reaction:

A reaction in which an insoluble solid (precipitate) is formed.

- **Example:** $AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$

F. Oxidation and Reduction (Redox Reactions):

- **Oxidation:** Gain of oxygen or loss of hydrogen or loss of electrons.
- **Reduction:** Loss of oxygen or gain of hydrogen or gain of electrons.
- **Example:** $ZnO + C \rightarrow Zn + CO$
 - ZnO loses oxygen $\rightarrow$ Reduced
 - C gains oxygen $\rightarrow$ Oxidised

G. Exothermic and Endothermic Reactions:

- **Exothermic:** Releases heat energy.
 - Example: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{Heat}$
- **Endothermic:** Absorbs heat energy.
 - Example: $2AgCl \rightarrow 2Ag + Cl_2$

4. Corrosion and Rancidity

A. Corrosion:

The gradual destruction of metals due to chemical reactions with their environment.

- **Rusting of Iron:** $4Fe + 3O_2 + 2xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O$
- **Prevention:** Painting, galvanisation, oiling, alloying, electroplating.

B. Rancidity:

The oxidation of fats and oils in food, giving it an unpleasant smell and taste.

- **Prevention:**
 - Adding antioxidants.
 - Packing in airtight containers.
 - Refrigeration.
 - Using nitrogen gas in food packets.

IMPORTANT QUESTIONS

Very Short Answer (1-2 marks):

1. Define a balanced chemical equation.
2. Identify the type of reaction: $2H_2 + O_2 \rightarrow 2H_2O$

3. What is the colour of ferrous sulphate crystals? What happens when they are heated?
4. Define oxidation and reduction with one example each.

****Short Answer (3 marks):****

1. Balance the following equations:
 - $\text{HNO}_3 + \text{Ca(OH)}_2 \rightarrow \text{Ca(NO}_3)_2 + \text{H}_2\text{O}$
 - $\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
2. Explain the following with chemical equations:
 - Thermal decomposition
 - Photolytic decomposition
 - Electrolytic decomposition
3. Differentiate between displacement and double displacement reactions with examples.

****Long Answer (5 marks):****

1. Describe with chemical equations all five types of chemical reactions. Give one example of each.
2. What is a redox reaction? Explain with the example of zinc oxide reacting with carbon.
3. Explain corrosion and rancidity with examples. How can they be prevented?

****Case-Based Question:****

Read the following passage and answer the questions:

"Silver chloride is a white powder that turns grey in sunlight. In photography, silver bromide is used."

- (i) Name the type of reaction taking place.
- (ii) Write the chemical equation for the decomposition of silver chloride.
- (iii) Why is silver bromide used in photography?
- (iv) What type of energy is required for this reaction?

CHAPTER 6: ACIDS, BASES AND SALTS

1. Acids and Bases – Definitions

****Acids:****

- Substances that furnish H^+ ions in aqueous solution.
- Sour taste, turn blue litmus red.
- ****Examples:**** HCl , H_2SO_4 , HNO_3 , CH_3COOH .

****Bases:****

- Substances that furnish OH^- ions in aqueous solution.

- Bitter taste, soapy touch, turn red litmus blue.
- **Examples:** NaOH, KOH, Ca(OH)₂, NH₄OH.

Alkalis: Water-soluble bases (e.g., NaOH, KOH).

Indicators:

- **Natural Indicators:** Litmus, Turmeric, Red cabbage juice.
- **Synthetic Indicators:** Methyl orange, Phenolphthalein.
- **Universal Indicator:** A mixture of indicators that gives different colours for different pH values.

2. Chemical Properties of Acids

A. Reaction with Metals:

Acid + Metal → Salt + Hydrogen gas

- **Example:** $2\text{HCl} + \text{Zn} \rightarrow \text{ZnCl}_2 + \text{H}_2\uparrow$

B. Reaction with Metal Carbonates/Hydrogen Carbonates:

Acid + Metal carbonate → Salt + CO₂ + H₂O

- **Example:** $2\text{HCl} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$
- **Test for CO₂:** Pass gas through lime water; it turns milky.

C. Reaction with Metal Oxides:

Acid + Metal oxide → Salt + H₂O

- **Example:** $2\text{HCl} + \text{CuO} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$

D. Reaction with Bases (Neutralisation):

Acid + Base → Salt + H₂O

- **Example:** $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

3. Chemical Properties of Bases

A. Reaction with Metals:

Base + Metal → Salt + H₂

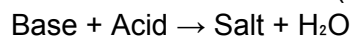
- **Example:** $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2\uparrow$

B. Reaction with Non-Metal Oxides:

Base + Non-metal oxide → Salt + H₂O

- **Example:** $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$

****C. Reaction with Acids (Neutralisation):****



4. pH Scale

The pH scale measures the concentration of H^+ ions in a solution.

- ****Range:**** 0 to 14
- ****pH < 7:**** Acidic
- ****pH = 7:**** Neutral
- ****pH > 7:**** Basic

****Importance of pH in Everyday Life:****

- ****Digestive System:**** HCl in stomach (pH ~1.5-2). Antacids neutralise excess acid.
- ****Tooth Decay:**** Bacteria produce acid which damages enamel (pH < 5.5). Toothpastes are basic.
- ****Soil pH:**** Plants require specific pH for growth.

5. Salts

Salts are ionic compounds formed by the neutralisation of an acid and a base.

****A. Types of Salts:****

- ****Normal Salts:**** Formed by complete neutralisation (e.g., NaCl).
- ****Acid Salts:**** Contain replaceable H^+ ions (e.g., NaHCO_3).
- ****Basic Salts:**** Contain replaceable OH^- ions (e.g., Pb(OH)Cl).

****B. pH of Salts:****

- Salt of strong acid and strong base: pH = 7 (e.g., NaCl).
- Salt of strong acid and weak base: pH < 7 (e.g., NH_4Cl).
- Salt of weak acid and strong base: pH > 7 (e.g., CH_3COONa).

****C. Salts and Their Uses:****

****Sodium Chloride (NaCl):****

- ****Preparation:**** Evaporation of sea water.
- ****Uses:**** Food; raw material for many chemicals.

****Sodium Hydroxide (NaOH):****

- ****Preparation:**** Chlor-alkali process: $2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2 + \text{Cl}_2$
- ****Uses:**** Soap, paper, detergents.

Bleaching Powder (CaOCl_2):

- **Preparation:** $\text{Ca(OH)}_2 + \text{Cl}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$
- **Uses:** Bleaching; disinfectant.

Baking Soda (NaHCO_3):

- **Preparation:** $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$
- **Uses:** Baking; antacid.

Washing Soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$):

- **Preparation:** $\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
- **Uses:** Glass, soap, water softening.

Plaster of Paris ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$):

- **Preparation:** $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + 1\frac{1}{2}\text{H}_2\text{O}$
- **Uses:** Casts; decorative items.

Water of Crystallisation:

The fixed number of water molecules present in one formula unit of a salt.

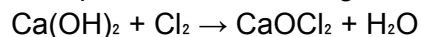
- $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$: Blue crystals (5 water molecules)
- $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$: 10 water molecules
- $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$: Gypsum (2 water molecules)
- $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$: Plaster of Paris ($\frac{1}{2}$ water molecule)

6. Important Reactions

Chlor-Alkali Process:



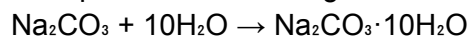
Preparation of Bleaching Powder:



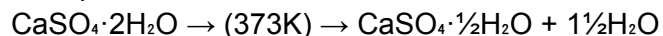
Preparation of Baking Soda:



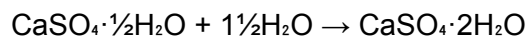
Preparation of Washing Soda:



Preparation of Plaster of Paris:



Setting of Plaster of Paris:



IMPORTANT QUESTIONS

****Very Short Answer (1-2 marks):****

1. Define an acid and a base according to Arrhenius theory.
2. What is the pH of pure water?
3. What is the chemical name of baking soda? Write its formula.
4. Why is Plaster of Paris stored in moisture-proof containers?

****Short Answer (3 marks):****

1. Explain the pH scale. What is its importance in everyday life?
2. Write the chemical equations for the reaction of zinc with sodium hydroxide.
3. What is water of crystallisation? Give two examples.
4. Differentiate between baking soda and washing soda with chemical formulae and uses.

****Long Answer (5 marks):****

1. Describe the preparation, properties, and uses of:
 - Sodium hydroxide
 - Bleaching powder
 - Baking soda
 - Washing soda
 - Plaster of Paris
2. Explain the reaction of acids with metals, metal carbonates, metal oxides, and bases.
3. What is the chlor-alkali process? Write the reaction and explain the uses of each product.

****Case-Based Question:****

Read the following and answer the questions:

"Rahul took a blue coloured solution in a test tube. He added a few drops of dilute sulphuric acid to it. The solution turned red."

- (i) What was the colour of the solution before adding acid?
- (ii) What is the nature of the solution?
- (iii) What would happen if he added a base to the red solution?
- (iv) What is the pH of the original solution?

CHAPTER 7: METALS AND NON-METALS

1. Physical Properties

Metals:

- **State:** Solid (except Hg-liquid).
- **Lustre:** Shiny.
- **Malleability:** Can be beaten into sheets.
- **Ductility:** Can be drawn into wires.
- **Conductivity:** Good conductors of heat and electricity.
- **Density:** Generally high.
- **Melting/Boiling Point:** Usually high.
- **Sonorous:** Yes.

Non-Metals:

- **State:** Solid, liquid (Br_2), gas.
- **Lustre:** Dull.
- **Malleability:** Brittle.
- **Ductility:** Not ductile.
- **Conductivity:** Poor conductors (except graphite).
- **Density:** Generally low.
- **Melting/Boiling Point:** Usually low.
- **Sonorous:** No.

Exceptions:

- Sodium and potassium: Soft.
- Mercury: Liquid at room temperature.
- Graphite (non-metal): Good conductor of electricity.
- Iodine (non-metal): Has lustre.

2. Chemical Properties of Metals

A. Reaction with Oxygen:

Metal + Oxygen $\rightarrow$ Metal oxide (Basic oxide)

- $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$
- $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$

Amphoteric Oxides: React with both acids and bases (ZnO , Al_2O_3).

B. Reaction with Water:

- **Na, K:** React violently with cold water.
 - $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2\uparrow$
- **Ca:** Reacts with cold water.
 - $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2\uparrow$

- **Mg:** Reacts with hot water/steam.
- **Al, Zn, Fe:** React with steam.
- **Cu, Ag, Au:** Do not react with water.

C. Reaction with Acids:

Metal + Dilute acid $\rightarrow$ Salt + $H_2\uparrow$

- $Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$
- **Note:** Cu, Ag, Au do not react with dilute HCl/ H_2SO_4 .

D. Reaction with Salt Solutions:

More reactive metal displaces less reactive metal from its salt solution.

- $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

3. Reactivity Series of Metals

The arrangement of metals in decreasing order of reactivity.

Complete Reactivity Series:

$K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au$

Important Rules:

- A more reactive metal can displace a less reactive metal from its salt solution.
- Metals above H can displace H from dilute acids.
- Metals below H cannot displace H from dilute acids.

4. Ionic Compounds

Formation:

- Metals lose electrons to form positive ions (cations).
- Non-metals gain electrons to form negative ions (anions).
- **Example: NaCl:** $Na \rightarrow Na^+ + e^-$; $Cl + e^- \rightarrow Cl^-$

Properties:

- **Physical state:** Solid at room temperature.
- **Melting/Boiling point:** High.
- **Solubility:** Usually soluble in water.
- **Conductivity:** Conduct electricity in molten state or in aqueous solution.

5. Extraction of Metals (Metallurgy)

A. Enrichment of Ore:

- Removal of gangue (impurities) from ore.

B. Extraction of Metal:

- **Conversion to Oxide:**

- **Calcination:** $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2\uparrow$
- **Roasting:** $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2\uparrow$

- **Reduction of Oxide:**

- **Smelting:** $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$
- **Electrolytic Reduction:** For reactive metals (Na, K, Mg, Al).

C. Refining of Metals:

- **Electrolytic Refining:** Impure metal is used as anode, pure metal as cathode.

Extraction of Highly Reactive Metals:

- **Aluminium (Bayer's Process + Hall-Heroult Process):**

- Bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$) $\rightarrow \text{Al}_2\text{O}_3 \rightarrow 2\text{Al}_2\text{O}_3 \rightarrow (\text{Electrolysis}) \rightarrow 4\text{Al} + 3\text{O}_2$

Extraction of Moderately Reactive Metals:

- **Iron (Blast Furnace):**

- $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

Extraction of Less Reactive Metals:

- **Mercury:** $\text{HgS (Cinnabar)} \rightarrow \text{HgO} \rightarrow 2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$

6. Corrosion of Metals

Corrosion: Gradual destruction of metals due to chemical reactions with environment.

Rusting of Iron:

- $4\text{Fe} + 3\text{O}_2 + 2x\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- **Conditions:** Presence of O_2 and moisture.
- **Prevention:** Painting, galvanisation, electroplating, alloying.

Tarnishing of Silver:

- $2\text{Ag} + \text{H}_2\text{S} \rightarrow \text{Ag}_2\text{S (black)} + \text{H}_2$

Green Coating on Copper:

- $2\text{Cu} + \text{CO}_2 + \text{H}_2\text{O} + \text{O}_2 \rightarrow \text{CuCO}_3 \cdot \text{Cu(OH)}_2$

7. Alloys

A homogeneous mixture of two or more metals, or a metal and a non-metal.

****Purpose of Alloying:****

- To increase strength and hardness.
- To improve corrosion resistance.
- To modify melting point.

****Common Alloys:****

- ****Steel:**** Fe + C; strong; construction.
- ****Stainless Steel:**** Fe + Cr + Ni; corrosion resistant; utensils.
- ****Brass:**** Cu + Zn; musical instruments.
- ****Bronze:**** Cu + Sn; statues.
- ****Solder:**** Pb + Sn; low melting point; soldering.
- ****Duralumin:**** Al + Cu + Mg + Mn; strong and light; aircraft.

IMPORTANT QUESTIONS

****Very Short Answer (1-2 marks):****

1. Why is sodium stored in kerosene oil?
2. What is an alloy? Give two examples.
3. Why do metals have high melting points?
4. What is the reactivity series of metals?
5. Why is copper used for electrical wires?
6. What is galvanisation?

****Short Answer (3 marks):****

1. Differentiate between metals and non-metals based on their physical and chemical properties.
2. Explain the extraction of aluminium from bauxite.
3. What is corrosion? Explain rusting of iron with chemical equation. How can it be prevented?
4. Compare the properties of ionic and covalent compounds.

****Long Answer (5 marks):****

1. Describe the steps involved in the extraction of a metal from its ore.
2. What is the reactivity series? How does it help in the extraction of metals?
3. Explain the formation of ionic compounds with the help of sodium chloride.
4. Describe the extraction of iron from its ore in a blast furnace.
5. What are alloys? Why are they made? Give examples and uses of common alloys.

****Case-Based Question:****

Read the following and answer the questions:

"A metal X is stored in kerosene oil. It reacts vigorously with water to produce a gas that burns with a pop sound."

- (i) Identify metal X.
- (ii) Write the reaction of X with water.
- (iii) Why is X stored in kerosene?
- (iv) What is the gas produced?

CHAPTER 8: CARBON AND ITS COMPOUNDS

1. Covalent Bonding in Carbon

****Covalent Bond:**** A bond formed by sharing of electrons between atoms.

****Why Carbon Forms Covalent Bonds:****

- Carbon has 4 valence electrons (electronic configuration: 2,4).
- It shares electrons with other atoms to form covalent bonds.

****Types of Covalent Bonds:****

- ****Single Bond:**** Sharing of one pair of electrons.
- ****Double Bond:**** Sharing of two pairs of electrons.
- ****Triple Bond:**** Sharing of three pairs of electrons.

****Properties of Covalent Compounds:****

- Low melting and boiling points.
- Poor conductors of electricity.
- Usually insoluble in water, soluble in organic solvents.

2. Versatile Nature of Carbon

****A. Catenation:****

- Ability of carbon atoms to bond with each other to form long chains, branched chains, or rings.
- ****Reason:**** Strong C–C bonds due to small size and tetravalency.

****B. Tetravalency:****

- Carbon has four valence electrons and can form four bonds.
- It can bond with elements like H, O, N, S, Cl, etc.

3. Hydrocarbons

Compounds made up of carbon and hydrogen only.

A. Saturated Hydrocarbons (Alkanes):

- All carbon atoms are bonded by single bonds.
- General formula: C_nH_{2n+2}
- **Examples:** Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8)

B. Unsaturated Hydrocarbons:

- **Alkenes:** Contain at least one double bond.
 - General formula: C_nH_{2n}
 - **Examples:** Ethene (C_2H_4), Propene (C_3H_6)
- **Alkynes:** Contain at least one triple bond.
 - General formula: C_nH_{2n-2}
 - **Examples:** Ethyne (C_2H_2), Propyne (C_3H_4)

4. Functional Groups and Homologous Series

Functional Group:

An atom or group of atoms that gives a compound its characteristic properties.

Common Functional Groups:

- **Alcohol:** $-OH$ ($R-OH$); Example: Ethanol (C_2H_5OH)
- **Aldehyde:** $-CHO$ ($R-CHO$); Example: Ethanal (CH_3CHO)
- **Carboxylic acid:** $-COOH$ ($R-COOH$); Example: Ethanoic acid (CH_3COOH)
- **Ketone:** $-CO-$ ($R-CO-R'$); Example: Propanone (CH_3COCH_3)
- **Haloalkane:** $-X$ ($R-X$); Example: Chloromethane (CH_3Cl)

Homologous Series:

A series of organic compounds with the same functional group and similar chemical properties, where each successive member differs by a CH_2 group.

Characteristics:

- Same general formula.
- Similar chemical properties.
- Gradual change in physical properties.

- Same functional group.

5. Nomenclature of Carbon Compounds

A. Alkanes (Suffix: -ane):

Methane (C₁), Ethane (C₂), Propane (C₃), Butane (C₄), Pentane (C₅), Hexane (C₆), Heptane (C₇), Octane (C₈), Nonane (C₉), Decane (C₁₀)

B. Alkenes (Suffix: -ene):

Ethene (C₂H₄), Propene (C₃H₆), Butene (C₄H₈)

C. Alkynes (Suffix: -yne):

Ethyne (C₂H₂), Propyne (C₃H₄), Butyne (C₄H₆)

D. Naming with Functional Groups:

- **Alcohols:** Replace -e of alkane with -ol

- CH₃CH₂OH: Ethanol

- **Carboxylic acids:** Replace -e with -oic acid

- CH₃COOH: Ethanoic acid

- **Aldehydes:** Replace -e with -al

- CH₃CHO: Ethanal

- **Ketones:** Replace -e with -one

- CH₃COCH₃: Propanone

E. Branched Chains:

- Identify the longest carbon chain.

- Number the chain so that substituents get the lowest numbers.

- Name substituents as alkyl groups.

- **Example:** 2-methylpropane

6. Chemical Properties of Carbon Compounds

A. Combustion:

Carbon compounds burn in oxygen to give CO₂, H₂O, heat, and light.

- CH₄ + 2O₂ → CO₂ + 2H₂O + Heat + Light

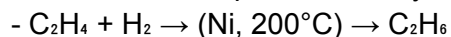
B. Oxidation:

Alcohols oxidise to carboxylic acids.

- C₂H₅OH + 2[O] → (Alkaline KMnO₄) → CH₃COOH + H₂O

****C. Addition Reaction (Unsaturated hydrocarbons):****

Unsaturated compounds add hydrogen in the presence of a catalyst to become saturated.



****D. Substitution Reaction (Saturated hydrocarbons):****

Hydrogen atoms in alkanes are replaced by other atoms/radicals.



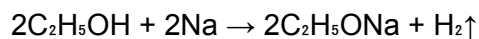
7. Ethanol (Ethyl Alcohol)

****Physical Properties:****

- Colourless liquid with pleasant smell.
- Miscible with water.
- Low boiling point (78°C).

****Chemical Properties:****

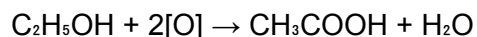
****A. Reaction with Sodium:****



****B. Dehydration (with Conc. H_2SO_4):****



****C. Oxidation:****



****Uses:****

- Beverages, solvent, fuel, antiseptic.

****Effects of Alcohol:****

- Depressant, liver damage, addiction.

****Denatured Alcohol:****

Ethanol mixed with poisonous substances to make it unfit for drinking.

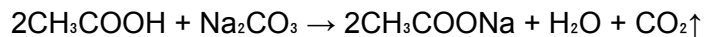
8. Ethanoic Acid (Acetic Acid)

****Physical Properties:****

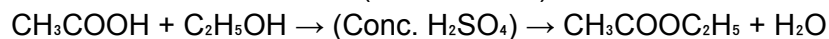
- Colourless liquid with pungent smell.
- Sour taste (vinegar contains 5-8% ethanoic acid).
- Melting point: 290K (forms glacial acetic acid).

****Chemical Properties:****

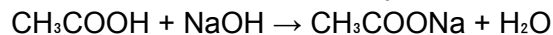
****A. Reaction with Sodium Carbonate:****



****B. Reaction with Ethanol (Esterification):****



****C. Reaction with Sodium Hydroxide:****



****Uses:****

- Vinegar, plastics, dyes, pharmaceuticals.

9. Soaps and Detergents

****Soaps:****

- Sodium or potassium salts of long-chain carboxylic acids.

****Preparation (Saponification):****

- Fat/Oil + NaOH $\rightarrow$ Soap + Glycerol

****Structure:****

- ****Hydrophilic Head:**** Ionic part (COONa).
- ****Hydrophobic Tail:**** Long hydrocarbon chain.

****Cleansing Action (Micelle Formation):****

1. Hydrophobic tails dissolve in grease/oil.
2. Hydrophilic heads remain in water.
3. Micelles trap grease and are washed away.

****Disadvantages of Soaps:****

- Do not work well in hard water.
- Form scum with Ca^{2+} and Mg^{2+} ions.

****Detergents:****

- Synthetic cleaning agents.
- ****Advantages:**** Work well in hard water; can be used in acidic solutions.
- ****Disadvantages:**** Non-biodegradable; cause water pollution.

IMPORTANT QUESTIONS

****Very Short Answer (1-2 marks):****

1. What is catenation?
2. Why does carbon form covalent bonds?
3. What is a homologous series?
4. Write the structural formula of ethanol and ethanoic acid.
5. Why is ethanol called an alcohol?
6. What is the difference between soap and detergent?

****Short Answer (3 marks):****

1. Explain the versatile nature of carbon.
2. What is a functional group? Name four functional groups with their general formulae.
3. Write the chemical equations for the following reactions:
 - Combustion of ethanol
 - Oxidation of ethanol
 - Dehydration of ethanol
 - Esterification
4. Explain the cleansing action of soap.

****Long Answer (5 marks):****

1. Describe the IUPAC nomenclature of hydrocarbons and functional groups.
2. Explain the chemical properties of carbon compounds with examples.
3. Write the reactions of ethanol and ethanoic acid with sodium, sodium carbonate, and ethanol.
4. Compare the properties of soaps and detergents.

****Case-Based Question:****

Read the following and answer:

"Ravi took a compound 'X' in a test tube. He added sodium bicarbonate to it. A gas was evolved which turned lime water milky. The compound 'X' on reaction with ethanol gave a sweet-smelling compound 'Y'."

- (i) Identify compound X.
- (ii) Write the reaction of X with sodium bicarbonate.
- (iii) Name the reaction between X and ethanol.
- (iv) Identify compound Y.

PHYSICS (UNIT III: NATURAL PHENOMENA & UNIT IV: EFFECTS OF CURRENT)

CHAPTER 9: LIGHT – REFLECTION AND REFRACTION

1. Light and Its Nature

****Light:**** A form of energy that causes the sensation of vision.

- Travels in a straight line.
- ****Speed in vacuum:**** 3×10^8 m/s.

2. Reflection of Light

****Reflection:**** Bouncing back of light from a smooth polished surface.

****Laws of Reflection:****

1. The incident ray, reflected ray, and normal at the point of incidence lie in the same plane.
2. The angle of incidence (i) equals the angle of reflection (r).

****Image Formation by Plane Mirror:****

- Virtual and erect.
- Same size as object.
- Laterally inverted.
- Distance of image = Distance of object.

3. Spherical Mirrors

****Types:****

- ****Concave Mirror:**** Reflecting surface curved inwards.
- ****Convex Mirror:**** Reflecting surface curved outwards.

****Important Terms:****

- ****Pole (P):**** Centre of the mirror's surface.
- ****Centre of Curvature (C):**** Centre of the sphere.
- ****Radius of Curvature (R):**** Distance from pole to centre of curvature.
- ****Focus (F):**** Point where parallel rays meet.
- ****Focal Length (f):**** Distance from pole to focus.
- ****Relation:**** $R = 2f$

****Image Formation by Concave Mirror:****

****Object at Infinity:****

- ****Image Position:**** At F.

- **Size:** Highly diminished.
- **Nature:** Real, inverted.

Object Beyond C:

- **Image Position:** Between F and C.
- **Size:** Diminished.
- **Nature:** Real, inverted.

Object at C:

- **Image Position:** At C.
- **Size:** Same size.
- **Nature:** Real, inverted.

Object Between C and F:

- **Image Position:** Beyond C.
- **Size:** Magnified.
- **Nature:** Real, inverted.

Object at F:

- **Image Position:** At infinity.
- **Size:** Highly magnified.
- **Nature:** Real, inverted.

Object Between F and P:

- **Image Position:** Behind mirror.
- **Size:** Magnified.
- **Nature:** Virtual, erect.

Image Formation by Convex Mirror:

- Always forms virtual, erect, and diminished images.
- Image is formed behind the mirror between P and F.

Sign Convention (Cartesian Sign Convention):

- **Object Distance (u):** Always negative.
- **Image Distance (v):** Positive for real; negative for virtual.
- **Focal Length (f):** Positive for convex; negative for concave.
- **Magnification (m):** Positive for virtual; negative for real.

Mirror Formula:

$$1/f = 1/v + 1/u$$

Magnification:

$$m = h'/h = -v/u$$

4. Uses of Spherical Mirrors

****Concave Mirrors:****

- Torch, searchlight, headlights.
- Shaving mirror.
- Doctor's head mirror.
- Solar cookers.

****Convex Mirrors:****

- Rear-view mirrors in vehicles.
- Security mirrors in shops.

5. Refraction of Light

****Refraction:**** Bending of light as it passes from one medium to another.

****Cause:**** Change in speed of light in different media.

****Laws of Refraction:****

1. The incident ray, refracted ray, and normal at the point of incidence lie in the same plane.
2. ****Snell's Law:**** $\sin i / \sin r = \text{constant (refractive index, } \mu \text{)}$.

****Refractive Index (μ):****

- $\mu = \sin i / \sin r = \text{Speed of light in vacuum} / \text{Speed of light in medium}$

****Absolute Refractive Index:****

- $\mu = c/v$

****Effects of Refraction:****

1. A coin in water appears raised.
2. A pencil in water appears bent.
3. A swimming pool appears shallower.

6. Spherical Lenses

****Types:****

- ****Convex Lens:**** Thicker at the centre; converges light.
- ****Concave Lens:**** Thinner at the centre; diverges light.

****Important Terms:****

- ****Optical Centre (O):**** Centre of the lens.
- ****Focus (F):**** Point where parallel rays meet.
- ****Focal Length (f):**** Distance from optical centre to focus.

****Image Formation by Convex Lens:****

****Object at Infinity:****

- ****Image Position:**** At F.
- ****Size:**** Highly diminished.
- ****Nature:**** Real, inverted.

****Object Beyond 2F:****

- ****Image Position:**** Between F and 2F.
- ****Size:**** Diminished.
- ****Nature:**** Real, inverted.

****Object at 2F:****

- ****Image Position:**** At 2F.
- ****Size:**** Same size.
- ****Nature:**** Real, inverted.

****Object Between F and 2F:****

- ****Image Position:**** Beyond 2F.
- ****Size:**** Magnified.
- ****Nature:**** Real, inverted.

****Object at F:****

- ****Image Position:**** At infinity.
- ****Size:**** Highly magnified.
- ****Nature:**** Real, inverted.

****Object Between F and O:****

- ****Image Position:**** Same side.
- ****Size:**** Magnified.
- ****Nature:**** Virtual, erect.

****Image Formation by Concave Lens:****

- Always forms virtual, erect, and diminished images.
- Image is formed on the same side as the object.

****Lens Formula:****

$$1/f = 1/v - 1/u$$

****Magnification:****

$$m = h'/h = v/u$$

****Power of a Lens (P):****

- $P = 1/f$ (f in metres)
- ****Unit:**** Dioptre (D).
- ****Convex Lens:**** $P > 0$ (positive).
- ****Concave Lens:**** $P < 0$ (negative).
- ****Combined Power:**** $P = P_1 + P_2 + P_3 + \dots$

IMPORTANT QUESTIONS

****Very Short Answer (1-2 marks):****

1. State the laws of reflection.
2. What is the relationship between radius of curvature and focal length of a spherical mirror?
3. Define refractive index.
4. What is the power of a lens?
5. What type of image is formed by a convex lens?

****Short Answer (3 marks):****

1. Derive the mirror formula.
2. Explain the rules for image formation by a concave mirror.
3. What is refraction? Explain Snell's law.
4. Differentiate between convex and concave lenses.
5. Give the sign convention for spherical lenses.

****Long Answer (5 marks):****

1. Describe the image formation by a concave mirror for different object positions.
2. What is the lens formula? Derive it.
3. Explain the working of a convex lens with ray diagrams.
4. Discuss the uses of spherical mirrors and lenses in daily life.

****Case-Based Question:****

Read the following and answer:

"A student uses a concave mirror to obtain a magnified image of an object on a screen."

- (i) Where should the object be placed?
- (ii) What is the nature of the image?
- (iii) What happens if the object is placed at the focus?
- (iv) Give one use of a concave mirror.

CHAPTER 10: THE HUMAN EYE AND THE COLOURFUL WORLD

1. Human Eye

****Parts of the Eye and Their Functions:****

****Cornea:****

- Transparent front part; allows light to enter; major refractive surface.

****Iris:****

- Coloured part; controls the amount of light entering.

****Pupil:****

- Black circular opening; regulates the amount of light.

****Lens:****

- Biconvex, transparent; focuses light on the retina; changes shape for accommodation.

****Ciliary Muscles:****

- Attached to the lens; change the shape of the lens.

****Retina:****

- Inner lining; contains light-sensitive cells (rods and cones).
- ****Rods:**** Vision in dim light.
- ****Cones:**** Colour vision.

****Optic Nerve:****

- Carries nerve impulses to the brain.

****Blind Spot:****

- Point where the optic nerve leaves the eye; no photoreceptor cells.

2. Defects of Vision and Their Correction

****A. Myopia (Short-sightedness):****

- ****Symptom:**** Near objects clear; far objects blurred.
- ****Cause:**** Elongated eyeball or high curvature of lens.
- ****Image:**** Forms in front of the retina.

- **Correction:** Concave lens.

B. Hypermetropia (Long-sightedness):

- **Symptom:** Far objects clear; near objects blurred.
- **Cause:** Shortened eyeball or weak lens.
- **Image:** Forms behind the retina.
- **Correction:** Convex lens.

C. Presbyopia:

- **Symptom:** Difficulty in viewing near objects.
- **Cause:** Loss of elasticity of the lens.
- **Correction:** Bifocal lens.

D. Astigmatism:

- **Symptom:** Distorted vision.
- **Cause:** Uneven curvature of the cornea.
- **Correction:** Cylindrical lens.

3. Power of Accommodation

Power of Accommodation: The ability of the eye lens to adjust its focal length.

- **Near Point:** 25 cm for a normal eye.
- **Far Point:** Infinity for a normal eye.

4. Refraction of Light Through a Prism

Prism: A wedge-shaped transparent object.

Angle of Deviation (δ):

- The angle between the incident ray and the emergent ray.

5. Dispersion of White Light

Dispersion: Splitting of white light into its constituent colours.

Sequence of Colours (VIBGYOR):

Violet, Indigo, Blue, Green, Yellow, Orange, Red

****Cause of Dispersion:****

- Different colours have different wavelengths.
- Violet has the shortest wavelength and is deviated the most.
- Red has the longest wavelength and is deviated the least.

****Recombination of Light:****

- A second prism placed inverted recombines the spectrum into white light.

6. Atmospheric Refraction and Scattering

****A. Atmospheric Refraction:****

- ****Twinkling of Stars:**** Due to varying refractive index of atmosphere.
- ****Advance Sunrise and Delayed Sunset:**** Due to refraction of sunlight.
- ****Apparent Position of Stars:**** Stars appear higher than their actual position.

****B. Scattering of Light:****

- ****Tyndall Effect:**** Scattering of light by colloidal particles.
- ****Blue Colour of the Sky:**** Blue light is scattered more than red.
- ****Red Colour of the Sun at Sunrise and Sunset:**** Blue light is scattered away; red light reaches our eyes.
- ****Red Colour of Danger Signals:**** Red light is scattered the least.

IMPORTANT QUESTIONS

****Very Short Answer (1-2 marks):****

1. What is the power of accommodation?
2. Define myopia. How is it corrected?
3. What is dispersion of light?
4. Why is the sky blue?
5. Why do stars twinkle?

****Short Answer (3 marks):****

1. Explain the structure and function of the human eye.
2. Describe the defects of vision and their corrections.
3. Explain the refraction of light through a prism.
4. Why is the sun red at sunrise and sunset?

****Long Answer (5 marks):****

1. Describe the human eye and its function with a labelled diagram.

2. Explain the phenomenon of dispersion of white light.
3. Discuss atmospheric refraction and its effects.
4. What is scattering of light? Explain the Tyndall effect.

****Case-Based Question:****

Read the following and answer:

"A person cannot see objects clearly beyond 2 metres. He uses a spectacle to see objects at infinity."

- (i) Name the defect of vision.
- (ii) What is the far point of this person?
- (iii) What type of lens is used for correction?
- (iv) Calculate the power of the lens required.

CHAPTER 11: ELECTRICITY

1. Electric Current

****Electric Current:**** The flow of electric charge.

- ****Symbol:**** I
- ****Unit:**** Ampere (A)
- ****Formula:**** $I = Q/t$

****Direction of Current:****

- ****Conventional Current:**** Positive to negative.
- ****Electron Flow:**** Negative to positive.

2. Electric Potential and Potential Difference

****Electric Potential (V):**** Work done in bringing a unit positive charge from infinity to a point.

****Potential Difference (V):**** Work done in moving a unit positive charge from one point to another.

- ****Formula:**** $V = W/Q$
- ****Unit:**** Volt (V)

3. Ohm's Law

Ohm's Law: At constant temperature, the current flowing through a conductor is directly proportional to the potential difference applied across its ends.

- **Mathematical Expression:** $V \propto I \rightarrow V = IR$

- Where: R = Resistance (Ohms, Ω)

Resistance (R):

- Opposition to the flow of current.

- **Unit:** Ohm (Ω)

4. Factors Affecting Resistance

Resistance depends on:

1. **Length (l):** $R \propto l$
2. **Area of Cross-section (A):** $R \propto 1/A$
3. **Material:** Different resistivity.
4. **Temperature:** R increases with temperature.

Resistivity (ρ):

- $R = \rho l/A$

- **Unit:** $\Omega \cdot m$

Conductors: Low resistivity.

Insulators: High resistivity.

Superconductors: Zero resistance at very low temperatures.

5. Combination of Resistors

A. Series Combination:

- **Current:** Same through all.

- **Voltage:** $V = V_1 + V_2 + V_3$

- **Equivalent Resistance:** $R_{eq} = R_1 + R_2 + R_3$

B. Parallel Combination:

- **Voltage:** Same across all.

- **Current:** $I = I_1 + I_2 + I_3$

- **Equivalent Resistance:** $1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3$

- **For two resistors:** $R_{eq} = (R_1 \times R_2) / (R_1 + R_2)$

6. Heating Effect of Electric Current

****Joule's Law of Heating:****

- $H = I^2 R t$

****Applications:****

1. Electric Iron, Heater, Toaster.
2. Electric Bulb (filament: Tungsten).
3. Fuse (protective device).

****Electric Power (P):****

- $P = VI = I^2 R = V^2 / R$

- ****Unit:**** Watt (W)

****Electric Energy (E):****

- $E = P \times t$

- ****Unit:**** Kilowatt-hour (kWh) or Unit

- ****1 kWh = 3.6×10^6 J****

7. Practical Applications

****A. Electric Fuse:****

- Thin wire of low melting point.
- Melts when current exceeds the rated value.

****B. Electric Bulb:****

- Filament: Tungsten (high melting point).
- Inert gas to prevent oxidation.

****C. Short Circuit:****

- Live and neutral wires touch directly.
- Causes high current, overheating, and fire.
- ****Prevention:**** Fuses, circuit breakers, proper insulation.

****D. Overloading:****

- More than the rated current flows through the circuit.
- ****Causes:**** Too many appliances, faulty wiring.
- ****Prevention:**** Proper wiring, appropriate fuses.

IMPORTANT QUESTIONS

****Very Short Answer (1-2 marks):****

1. Define electric current.
2. State Ohm's law.
3. What is resistivity?
4. Define one kilowatt-hour.
5. What is the function of a fuse?

****Short Answer (3 marks):****

1. Derive the expression for equivalent resistance in series and parallel.
2. Explain Joule's law of heating.
3. What factors affect resistance? How?
4. Differentiate between series and parallel combinations.

****Long Answer (5 marks):****

1. Explain Ohm's law. Describe an experiment to verify it.
2. What is the heating effect of current? Give its applications.
3. Describe the factors affecting resistance.
4. Explain the combination of resistors in series and parallel with formulae.

****Case-Based Question:****

Read the following and answer:

"A student connects three resistors of 2Ω , 3Ω , and 5Ω in series to a 10V battery."

- (i) What is the total resistance?
- (ii) What is the current flowing through the circuit?
- (iii) What is the voltage across each resistor?
- (iv) What would happen if the resistors were connected in parallel?

CHAPTER 12: MAGNETIC EFFECTS OF ELECTRIC CURRENT

1. Magnetic Field and Field Lines

****Magnetic Field:****

- The region around a magnet where its influence is felt.

****Properties of Magnetic Field Lines:****

1. Originate from the north pole and end at the south pole.

2. Form closed loops.
3. Never intersect each other.
4. Crowded near poles (strong field).
5. Tangent at any point gives the direction of the field.

2. Magnetic Field due to a Current-Carrying Conductor

A. Straight Conductor:

- **Rule:** Right-Hand Thumb Rule.
 - Thumb: Direction of current.
 - Fingers: Direction of magnetic field.

B. Circular Loop:

- **Rule:** Right-Hand Rule for Loop.
 - Fingers curl in the direction of current.
 - Thumb points in the direction of magnetic field.

C. Solenoid:

- Long coil of wire with many turns.
- **Magnetic Field:** Similar to a bar magnet; strong and uniform inside.

Electromagnet:

- A solenoid with an iron core.
- Becomes magnetic when current flows; loses magnetism when current stops.

3. Force on a Current-Carrying Conductor

Fleming's Left-Hand Rule:

- **Thumb:** Force.
- **Forefinger:** Magnetic Field.
- **Middle Finger:** Current.

Formula:

- $F = BIL \sin\theta$

Applications:

- Electric motor, galvanometer.

4. Electric Motor

****Principle:**** A current-carrying conductor placed in a magnetic field experiences a force.

****Construction:****

1. ****Armature:**** Rectangular coil.
2. ****Magnet:**** Provides magnetic field.
3. ****Split Rings (Commutator):**** Reverses direction of current.
4. ****Brushes:**** Connect commutator to power supply.

****Working:****

1. Current flows through the coil.
2. Forces act on the two sides.
3. The coil rotates.
4. Commutator reverses current after every half rotation.

****Applications:****

- Electric fans, washing machines, refrigerators.

5. Electromagnetic Induction

****Electromagnetic Induction:**** The process of generating induced current by changing magnetic flux.

****Fleming's Right-Hand Rule:****

- ****Thumb:**** Motion.
- ****Forefinger:**** Magnetic Field.
- ****Middle Finger:**** Current (Induced).

****Factors Affecting Induced Current:****

1. Speed of motion.
2. Strength of magnetic field.
3. Number of turns in coil.

6. Electric Generator

****Principle:**** Electromagnetic induction.

****A. AC Generator:****

- ****Parts:**** Armature, magnets, slip rings, brushes.

- Produces alternating current (50 Hz in India).

****B. DC Generator:****

- Uses a commutator.
- Produces direct current.

7. Alternating Current (AC) vs Direct Current (DC)

****AC:****

- Direction changes (alternates).
- Magnitude varies.
- Frequency: 50 Hz (in India).
- Easy transmission (can be stepped up/down).
- Less energy loss.

****DC:****

- Flows in one direction.
- Magnitude constant.
- Difficult transmission.

****Advantages of AC over DC:****

- Can be transformed to high voltage.
- Less energy loss in transmission.
- Can be converted to DC easily.

8. Domestic Electric Circuits

****Wiring in India:****

1. ****Three Wires:****

- ****Live Wire (Red/Brown):**** Carries current.
- ****Neutral Wire (Black/Blue):**** Completes the circuit.
- ****Earth Wire (Green/Yellow):**** Safety wire.

2. ****Voltage:**** 220 V (AC, 50 Hz).

3. ****Fuse:**** Connected in series with live wire.

4. ****Earthing:**** Provides a low-resistance path.

5. ****Parallel Connection:**** Each appliance gets the same voltage.

****Short Circuit:****

- Live and neutral wires touch → Low resistance → High current → Fire.

****Overloading:****

- Drawing more current than the circuit can handle.
- ****Prevention:**** Proper wiring, fuses, circuit breakers.

IMPORTANT QUESTIONS

****Very Short Answer (1-2 marks):****

1. What is a magnetic field?
2. State Fleming's left-hand rule.
3. What is electromagnetic induction?
4. What is the difference between AC and DC?
5. What is the function of an earth wire?

****Short Answer (3 marks):****

1. Explain the magnetic field due to a current-carrying solenoid.
2. Describe the working of an electric motor.
3. What is electromagnetic induction? Explain Faraday's experiment.
4. Explain the working of an AC generator.

****Long Answer (5 marks):****

1. Describe the magnetic field of a current-carrying straight conductor, circular loop, and solenoid.
2. Explain the construction and working of an electric motor.
3. What is electromagnetic induction? Explain the principle and working of a generator.
4. Compare AC and DC. Why is AC preferred for transmission?
5. Explain the domestic electric circuit with a diagram.

****Case-Based Question:****

Read the following and answer:

"A student observes that when a magnet is moved towards a coil connected to a galvanometer, the needle deflects."

- (i) What is this phenomenon called?
- (ii) What would happen if the magnet is moved away?
- (iii) What would happen if the magnet is kept stationary?
- (iv) State the rule used to find the direction of induced current.

UNIT V: NATURAL RESOURCES

CHAPTER 13: OUR ENVIRONMENT

1. Ecosystem

****Ecosystem:**** A self-contained unit consisting of living organisms and their non-living environment.

****Components of an Ecosystem:****

****A. Biotic (Living):****

1. ****Producers:**** Green plants, algae, cyanobacteria.
2. ****Consumers:****
 - ****Primary Consumers (Herbivores):**** Feed on plants.
 - ****Secondary Consumers (Carnivores):**** Feed on herbivores.
 - ****Tertiary Consumers:**** Feed on secondary consumers.
 - ****Omnivores:**** Feed on both plants and animals.
3. ****Decomposers:**** Bacteria, fungi; break down dead organic matter.

****B. Abiotic (Non-Living):****

- Air, water, soil, sunlight, temperature, minerals.

2. Food Chains and Food Webs

****Food Chain:****

- A sequence of organisms in which each organism feeds on the previous one.
- ****Example:**** Grass → Grasshopper → Frog → Snake → Eagle

****Trophic Levels:****

1. ****First:**** Producers (Grass)
2. ****Second:**** Primary Consumers (Grasshopper)
3. ****Third:**** Secondary Consumers (Frog)
4. ****Fourth:**** Tertiary Consumers (Snake)

****Food Web:****

- Interconnected food chains.
- More complex and stable.

****Energy Flow:****

- Energy flows in one direction: Sun → Producers → Consumers → Decomposers.

- **10% Law:** Only 10% of energy at one trophic level is transferred to the next.

3. Ozone Layer Depletion

Ozone (O₃):

- A molecule of three oxygen atoms.
- **Location:** Stratosphere.
- **Function:** Absorbs harmful UV radiation.

Depletion of Ozone Layer:

- **Cause:** Chlorofluorocarbons (CFCs).
- **Mechanism:** CFCs release chlorine atoms; chlorine reacts with ozone.
- **Effects:** Increased UV radiation → Skin cancer, cataracts.
- **Prevention:** Montreal Protocol (1987) to phase out CFCs.

4. Waste Production and Solutions

Types of Waste:

A. Biodegradable Wastes:

- Can be decomposed by microorganisms.
- **Examples:** Vegetable peels, paper, animal dung.

B. Non-Biodegradable Wastes:

- Cannot be decomposed by microorganisms.
- **Examples:** Plastics, glass, metals, pesticides.

Solutions:

1. **Recycling:** Converting waste into reusable materials.
2. **Composting:** Converting organic waste into manure.
3. **Incineration:** Burning waste.
4. **Landfilling:** Burying waste.
5. **Waste Segregation:** Separating waste types.
6. **Reduce, Reuse, Recycle (3Rs).**

5. Environmental Problems

A. Pollution:

- **Air Pollution:** Vehicular emissions, industrial smoke.
- **Water Pollution:** Industrial waste, sewage, agricultural runoff.
- **Soil Pollution:** Chemical fertilisers, pesticides, industrial waste.

B. Global Warming:

- **Cause:** Increase in greenhouse gases (CO₂, methane).
- **Effects:** Rising temperatures, melting glaciers, climate change.
- **Prevention:** Reduce emissions, use renewable energy, plant trees.

C. Acid Rain:

- **Cause:** SO₂ and NO_x from burning fossil fuels.
- **Effects:** Acidification of lakes, damage to crops and buildings.
- **Prevention:** Reduce SO₂ and NO_x emissions.

6. Biodegradable and Non-Biodegradable Substances

Biodegradable:

- Can be decomposed by microorganisms.
- Recycles nutrients.
- **Examples:** Paper, vegetable waste, leather, cotton.

Non-Biodegradable:

- Cannot be decomposed.
- Accumulates and causes pollution.
- **Examples:** Plastics, glass, metals, pesticides.

Problems with Non-Biodegradable Wastes:

- Accumulate in the environment.
- May be toxic.
- Enter food chains (bioaccumulation).
 - **DDT:** Accumulates in organisms and causes health problems.

IMPORTANT QUESTIONS

Very Short Answer (1-2 marks):

1. Define an ecosystem.
2. What is a food chain? Give an example.
3. What is the 10% law?
4. What is ozone depletion? What causes it?
5. Define biodegradable and non-biodegradable wastes.

****Short Answer (3 marks):****

1. Differentiate between producers, consumers, and decomposers.
2. Explain the 10% law of energy transfer.
3. What are the effects of ozone depletion?
4. How can we manage waste effectively?

****Long Answer (5 marks):****

1. Describe the structure and function of an ecosystem.
2. Explain energy flow in an ecosystem with a food chain.
3. Discuss the environmental problems caused by pollution.
4. What is ozone depletion? How can it be prevented?
5. Differentiate between biodegradable and non-biodegradable wastes.

****Case-Based Question:****

Read the following and answer:

"A farmer uses chemical pesticides and fertilisers in his fields. The nearby pond shows a decrease in aquatic life."

- (i) What type of pollution is occurring?
- (ii) How are pesticides affecting aquatic life?
- (iii) What is biomagnification?
- (iv) Suggest alternatives to chemical pesticides and fertilisers.

IMPORTANT DIAGRAMS TO PRACTISE

1. Human Digestive System
2. Human Respiratory System
3. Structure of Heart
4. Structure of Nephron
5. Neuron
6. Human Brain
7. Male Reproductive System
8. Female Reproductive System
9. Flower (Parts)
10. Reflection/Refraction Ray Diagrams
11. Human Eye
12. Electric Motor
13. Electric Generator
14. Food Chain/Web
15. Ozone Layer

KEY FORMULAE

Physics:

1. **Ohm's Law:** $V = IR$
2. **Resistance:** $R = \rho l/A$
3. **Power:** $P = VI = I^2R = V^2/R$
4. **Energy:** $E = Pt$
5. **Mirror Formula:** $1/f = 1/v + 1/u$
6. **Lens Formula:** $1/f = 1/v - 1/u$
7. **Magnification:** $m = h'/h = -v/u$
8. **Power of Lens:** $P = 1/f$ (in metres)
9. **Refractive Index:** $\mu = \sin i / \sin r = c/v$

Chemistry:

1. **pH = $-\log[H^+]$
2. **Valency:** Number of electrons lost/gained/shared.

Biology:

1. **Photosynthesis:** $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$
2. **Aerobic Respiration:** $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy}$
3. **Anaerobic Respiration:** $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + \text{Energy (or Lactic acid)}$

QUICK REVISION TIPS

1. Make flashcards for important definitions and formulas.
2. Practise drawing diagrams without labels and then label them.
3. Solve previous year questions to understand the exam pattern.
4. Group related topics (e.g., all equations, all hormones).
5. Read the NCERT textbook thoroughly.
6. Practise numerical problems for Physics and Chemistry.
7. Use mnemonics for memorising lists:
 - **VIBGYOR:** Violet, Indigo, Blue, Green, Yellow, Orange, Red.
 - **Reactivity Series:** Potassium Sodium Calcium Magnesium Aluminium Zinc Iron Lead Hydrogen Copper Mercury Silver Gold

Best of luck for your CBSE Class 10 Science examination!